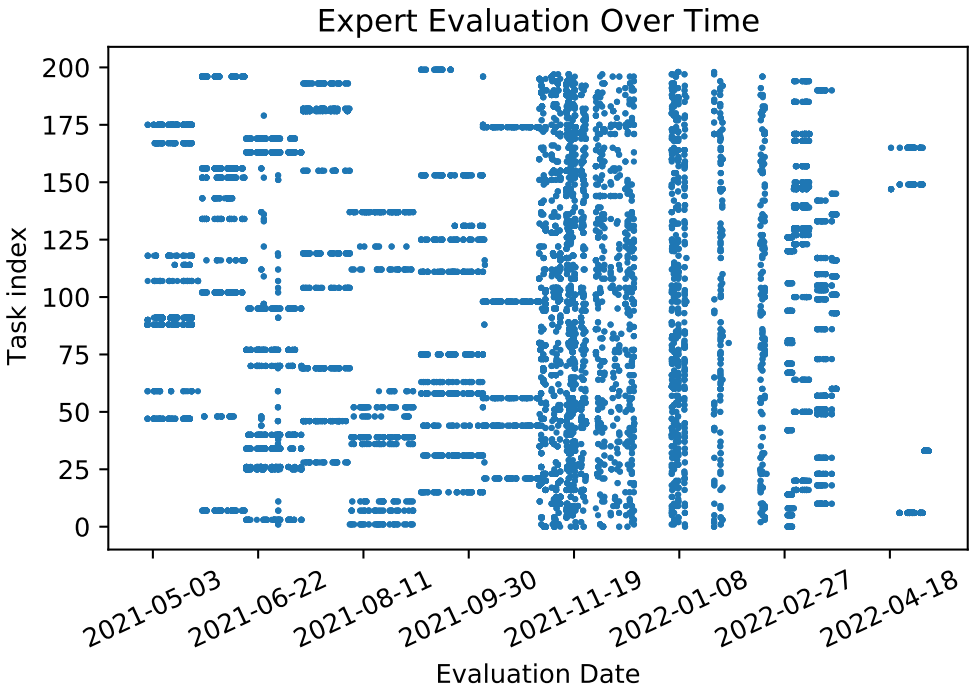

Figure App.8: **Timeline of expert evaluation over the course of BIG-bench.** Early evaluation was performed over restricted sets of the benchmark. Later testing shifted to more uniformly subsampled evaluation across the full benchmark.

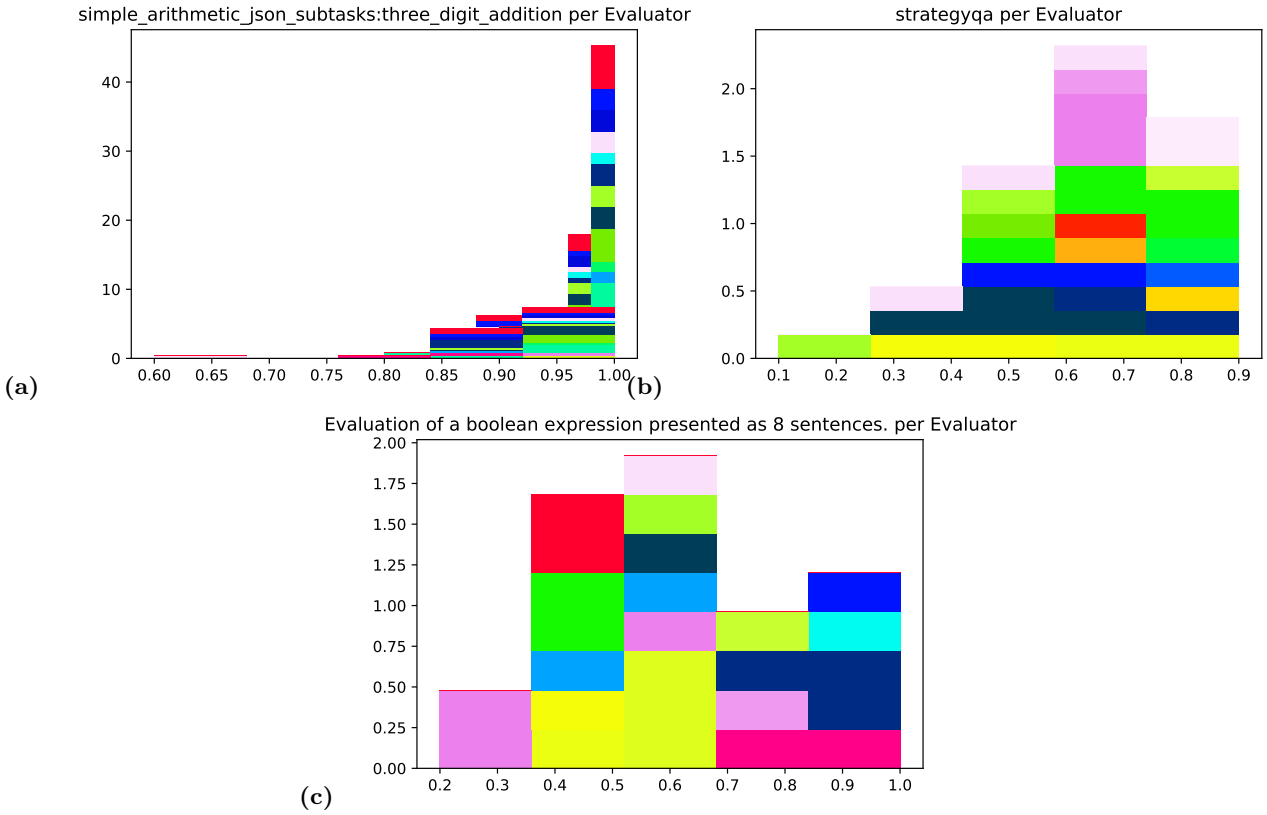

Figure App.9: **Representative histograms of expert evaluators performance** on an **(a)** easy, **(b)** intermediate, and **(c)** hard task. Identical colors represent repeated evaluation by the same evaluator.

References

Wikiquote, russian proverbs. URL https://ru.wikiquote.org/wiki/%D0%A0%D1%83%D1%81%D1%81%D0%BA%D0%B8%D0%B5_%D0%BF%D0%BE%D1%81%D0%BB%D0%BE%D0%B2%D0%B8%D1%86%D1%8B. (cited on p. 32)

Abubakar Abid, Maheen Farooqi, and James Zou. Persistent anti-Muslim bias in large language models, 2021. URL https://arxiv.org/abs/2101.05783. (cited on pp. 29 and 36)

Joshua Ackerman and George Cybenko. A survey of neural networks and formal languages, 2020. URL https://arxiv.org/abs/2006.01338. (cited on p. 31)

Aishwarya Agrawal, Jiasen Lu, Stanislaw Antol, Margaret Mitchell, C. Lawrence Zitnick, Dhruv Batra, and Devi Parikh. VQA: Visual question answering, 2015. URL https://arxiv.org/abs/1505.00468. (cited on p. 34)

Akshay Agrawal, Shane Barratt, and Stephen Boyd. Learning convex optimization models, 2020. URL https://arxiv.org/abs/2006.04248. (cited on p. 38)

Scott Alexander. A very unlikely chess game, 2020. URL https://slatestarcodex.com/2020/01/06/a-very-unlikely-chess-game/. (cited on pp. 29 and 30)

Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. Asking clarifying questions in open-domain information-seeking conversations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, New York, NY, USA, July 2019. Association for Computing Machinery. doi: 10.1145/3331184.3331265. URL https://dl.acm.org/doi/10.1145/3331184.3331265. (cited on p. 40)

Miltiadis Allamanis, Pankajan Chanthirasegaran, Pushmeet Kohli, and Charles Sutton. Learning continuous semantic representations of symbolic expressions, 2016. URL https://arxiv.org/abs/1611.01423. (cited on p. 29)

Miltiadis Allamanis, Earl T. Barr, Premkumar Devanbu, and Charles Sutton. A survey of machine learning for big code and naturalness. *ACM Comput. Surv.*, 51(4), July 2018. doi: 10.1145/3212695. URL https://doi.org/10.1145/3212695. (cited on p. 37)

Uri Alon, Shaked Brody, Omer Levy, and Eran Yahav. code2seq: Generating sequences from structured representations of code, 2018. URL https://arxiv.org/abs/1808.01400. (cited on p. 30)

Uri Alon, Roy Sadaka, Omer Levy, and Eran Yahav. Structural language models of code. In Hal Daumé III and Aarti Singh (eds.), *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 245–256. PMLR, 13–18 July 2020. URL https://proceedings.mlr.press/v119/alon20a.html. (cited on p. 37)

Miriam Amin and Manuel Burghardt. A survey on approaches to computational humor generation. In *Proceedings of the 4th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature*, pp. 29–41, Online, December 2020. International Committee on Computational Linguistics. URL https://aclanthology.org/2020.latechclfl-1.4. (cited on p. 38)

Aida Amini, Saadia Gabriel, Shanchuan Lin, Rik Koncel-Kedziorski, Yejin Choi, and Hannaneh Hajishirzi. MathQA: Towards interpretable math word problem solving with operation-based formalisms. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 2357–2367, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1245. URL https://aclanthology.org/N19-1245. (cited on pp. 31 and 37)

Prithviraj Ammanabrolu, William Broniec, Alex Mueller, Jeremy Paul, and Mark O. Riedl. Toward automated quest generation in text-adventure games, 2019. URL https://arxiv.org/abs/1909.06283. (cited on p. 29)

Prithviraj Ammanabrolu, Wesley Cheung, Dan Tu, William Broniec, and Mark O. Riedl. Bringing stories alive: Generating interactive fiction worlds, 2020. URL https://arxiv.org/abs/2001.10161. (cited on p. 29)

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety, 2016. URL https://arxiv.org/abs/1606.06565. (cited on p. 40)

Brandon Amos and J. Zico Kolter. Optnet: Differentiable optimization as a layer in neural networks, 2017. URL https://arxiv.org/abs/1703.00443. (cited on p. 38)

Philip W. Anderson. More is different. *Science*, 177(4047):393–396, 1972. doi: 10.1126/science.177.4047.393. URL https://www.science.org/doi/abs/10.1126/science.177.4047.393. (cited on p. 4)

Issa Annamoradnejad and Gohar Zoghi. ColBERT: Using BERT sentence embedding for humor detection, 2020. URL https://arxiv.org/abs/2004.12765. (cited on pp. 38 and 39)

Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4623–4637, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.421. URL https://aclanthology.org/2020.acl-main.421. (cited on p. 33)